YASH RANJAN

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EDUCATION

University of Florida *06/2024 - 12/2027 (expected)*
- PhD - Computer Science
Advisors: Dr. Sanjay Ranka, Dr. Anand Rangarajan

University of Florida *08/2022 - 05/2024*
- M.Sc.(Hons.) - Computer Science

BITS Pilani, India *08/2016 - 06/2020*
- B.E.(Hons.) Computer Science

WORK EXPERIENCE

Research Assistant, University of Florida *05/2024 - present*
- Built a Transformer-based generative world model for multi-agent traffic simulation (vehicles, pedestrians, signal phases, scene geometry), with a data harmonization framework that trains across intersections; improved minADE by 36% over single-scene training with strong zero-shot generalization to held-out intersections. [**BigData '26**, **arXiv '26**]
- Developing a constrained-autoencoder framework (hard linear equality constraints via Lagrange-multiplier units) for scientific data compression and latent-space surrogate simulation, using conservation constraints (density, flow, temperature moments) to control long-horizon rollout drift.
- Developed a VAE-based trajectory prediction model combining curvilinear coordinates with intersection controls, improving displacement error by 17% over baseline. [**IV '24**]
- Designed a simulation-in-the-loop framework for evaluating trajectory prediction models on multi-agent simulation, with traffic-engineering metrics for unsafe deceleration, lane following, and red-light violations. [**ITSC '25**, **PAKDD '25**]
- Maintained and operated the end-to-end computer vision data pipeline for traffic-intersection video analytics, configuring YOLOv7 object detection and DeepSORT tracking to deliver high-quality trajectory data for near-miss conflict analysis, with over 99% uptime. [**ICSTTE '25**]

Software Intern, Nutanix, San Jose, CA *05/2023 - 08/2023*
- Engineered an efficient messaging bus for security policy framework to send and receive security policies between hypervisor and management VM and stress tested it to 1M+ policies, reducing latency by over 15%.
- Created a front-end page in React with dynamic page refresh to show the security policies applied on containers running on the hypervisor with controls for adding subnet masks and IP addresses.

Member of Technical Staff, VMware, Bangalore, India *07/2020 - 07/2022*
- Worked on the Edge-Routing team developing routing protocols for the open-source routing framework Free Range Routing (FRRouting).
- Designed and implemented data structures for transaction-driven asynchronous CRUD operations for management of network routing information.
- Contributed features such as commit-rollback (reversing the transaction), asynchronous read operations (for fetching large operational data without delaying other transactions), and wildcard query resolution (similar to *find* command in GNU).

Undergraduate Researcher, BITS Pilani, India *08/2018 - 05/2019*
- Built a scalable Apache Spark pipeline for community detection (modularity 0.91), feature extraction, and event prediction on a 64K-node, 1.5M-edge social network, with an ensemble of Gradient Boosting, Random Forest, and SVM classifiers achieving 82% test accuracy. [**Arab J Sci Eng '20**]

PREPRINTS AND PUBLICATIONS

[1] **Ranjan Y**, Sengupta R, Rangarajan A, Ranka S. *PhaseShift: Topology-Aware Data Harmonization and Model Consolidation Across Signalized Intersections* 2026 IEEE International Conference on Big Data (BigData) https://arxiv.org/abs/2608.25275

[2] **Ranjan Y**, Sengupta R, Rangarajan A, Ranka S. *A Generative Model for Closed-Loop Microsimulation of Signalized Intersections* (Under review) https://arxiv.org/abs/2606.23588

[3] **Ranjan Y**, Sengupta R, Rangarajan A, Ranka S. *Enactor: From Traffic Simulators to Surrogate World Models.* 2026 International Conference on Vehicle Technology and Intelligent Transport Systems https://arxiv.org/abs/2603.18266

[4] **Ranjan Y**, Sengupta R, Rangarajan A, Ranka S. *IntTrajSim: Trajectory Prediction for Simulating Multi-Vehicle driving at Signalized Intersections.* 2025 IEEE International Conference on Intelligent Transportation Systems https://arxiv.org/abs/2506.08957

[5] **Ranjan Y**, Sengupta R, Rangarajan A, Ranka S. *Evaluating Generative Vehicle Trajectory Models for Traffic Intersection Dynamics.* 2025 Pacific Asia Conference on Knowledge Discovery and Data Mining (Special Session) https://arxiv.org/abs/2506.08963

[6] Wu A, **Ranjan Y**, Sengupta R, Rangarajan A, Ranka S. *A Data-driven Approach for Probabilistic Traffic Prediction and Simulation at Signalized Intersections.* 2024 IEEE Intelligent Vehicles Symposium (IV) https://doi.org/10.1109/IV55156.2024.10588424

[7] Sengupta R, Yousefzadeh N, Sanghvi M, **Ranjan Y**, Rangarajan A, Ranka S, Karnati Y, Dilmore J, Patel T, Casburn R. *BigSUMO: A Scalable Framework for Big Data Traffic Analytics and Parallel Simulation.* 2025 International Conference on Modeling, Simulation & Intelligent Computing (MoSICom) https://arxiv.org/abs/2510.19894

[8] Fleischer J, **Ranjan Y**, Banerjee T, Li MT, Rangarajan A, Ranka S. *Video-Based Machine Learning Approaches for Quantifying Pedestrian Safety in Traffic Intersections.* International Conference on SmartRail, Traffic and Transportation Engineering (ICSTTE 2025) https://doi.org/10.1007/978-981-92-3566-7_49

[9] Rajita B.S.A.S, **Ranjan Y**, Umesh C.T, Panda S. *Spark-Based Parallel Method for Prediction of Events.* Arab J Sci Eng 45, 3437–3453 (2020) https://doi.org/10.1007/s13369-020-04437-2

TECHNICAL EXPERTISE

Languages	Python, C, C++, Go, Java, JavaScript
Architecture	Transformer, VAE, Diffusion Model
Machine Learning	PyTorch, Apache Spark, YOLOv7, DeepSORT
Tools	SUMO, TensorBoard, Weights & Biases, GitHub, Docker